H524 Introduction to Biostatistics

Week 7: Nonparametric Methods

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Outline

What We've Learned So Far

Parametric Methods (Weeks 1-6):

- One-sample t-test
- Two-sample t-test (independent and paired)
- ANOVA for comparing multiple groups
- Correlation and regression

Key Assumptions of Parametric Tests:

1. Data follow a specific distribution (usually normal)
2. Parameters (μ, σ) characterize the population
3. Appropriate sample size for CLT
4. Equal variances (for some tests)

Question: What if these assumptions are violated?

When Parametric Assumptions Fail

Common violations:

- **Skewed data:** Income, hospital stay length, survival times
- **Outliers:** Extreme values that distort the mean
- **Ordinal data:** Pain scales (1-10), disease severity ratings
- **Small samples:** CLT doesn't apply, normality uncertain
- **Unequal variances:** Heteroscedasticity

Example: Pain Scale Data

- Pain rated 1-10 (ordinal)
- Difference between 2 and 3 may not equal difference between 8 and 9
- Mean may not be meaningful
- Normal distribution assumption questionable

Solution: Nonparametric methods (also called distribution-free methods)

What Are Nonparametric Methods?

Definition: Statistical procedures that make minimal assumptions about the underlying population distribution

Advantages:

- No normality assumption required
- Robust to outliers
- Work with ordinal data
- Valid for small sample sizes
- Often based on ranks (relative ordering)

Disadvantages:

- Less powerful than parametric tests when assumptions hold
- May lose information by using ranks instead of raw values
- Sometimes harder to interpret

Key Insight: Use ranks rather than actual data values

Ranks: The Foundation of Nonparametric Methods

Ranking: Ordering observations from smallest to largest

Example: Patient recovery times (days)

Patient	Recovery Time	Rank	Notes
A	3 days	1	Fastest
B	5 days	2	
C	7 days	3	
D	9 days	4	
E	45 days	5	Outlier!

Key observations:

- Mean = 13.8 days (heavily influenced by outlier)
- Median = 7 days (rank 3, not affected by outlier)
- Ranks preserve ordering but reduce outlier impact
- Rank 5 could be 45 days or 450 days - doesn't matter!

Handling Tied Ranks

What if observations have the same value?

Example: Pain scores

Patient	Pain Score	Naive Rank	Average Rank
A	2	1	1
B	4	2	2.5
C	4	3	2.5
D	6	4	4
E	7	5	5

Tied rank rule: Assign the average of the ranks that would have been assigned

- Patients B and C both scored 4
- Would have received ranks 2 and 3
- Both get rank $(2+3)/2 = 2.5$

The Sign Test: Simplest Nonparametric Test

Purpose: Test whether population median equals a hypothesized value

When to use:

- One-sample location test
- Small sample sizes
- Non-normal data
- Outliers present
- Only care about direction of difference

How it works:

1. Subtract hypothesized median from each observation
2. Count number of positive differences (call this B^+)
3. Count number of negative differences (call this B^-)
4. Ignore zeros (ties with hypothesized median)
5. If null hypothesis true: B^+ and B^- should be roughly equal

Test statistic: $B = \min(B^+, B^-)$

Sign Test: Hypotheses

Null hypothesis: H_0: Population median $= M_0$

Alternative hypotheses:

- **Two-sided:** H_A: Population median $\neq M_0$
- **One-sided (upper):** H_A: Population median $> M_0$
- **One-sided (lower):** H_A: Population median $< M_0$

Distribution under H_0:

- If true median $= M_0$, each observation has 50% chance of being above or below M_0
- Number of positives follows Binomial$(n, p = 0.5)$
- Can use binomial distribution to calculate p-values

Decision rule: Reject H_0 if p-value $< \alpha$

Sign Test Example

Research Question: Is median recovery time different from 6 days?

Data: Recovery times (days) for 9 patients:
8, 5, 7, 10, 4, 9, 3, 12, 6

Hypotheses:

- H_0: Median $=$ 6 days
- H_A: Median $\neq$ 6 days

Calculate differences from 6:

Observation	Value	Difference	Sign
1	8	$+2$	$+$
2	5	-1	-
3	7	$+1$	$+$
4	10	$+4$	$+$
5	4	-2	-
6	9	$+3$	$+$

Sign Test Example (continued)

Count signs:

- $B^+ = 5$ (positive differences)
- $B^- = 3$ (negative differences)
- $n = 8$ (excluding the tie)

Test statistic: $B = \min(5, 3) = 3$

P-value calculation:

- Under H_0: Number of positives $\sim \text{Binomial}(8, 0.5)$
- Two-sided test: $P = 2 \times \Pr(X \leq 3) = 2 \times 0.363 = 0.726$
- In R: `2 * pbinom(3, 8, 0.5) = 0.726`

Conclusion:

- p-value $= 0.726 > 0.05$
- Fail to reject H_0
- No evidence that median differs from 6 days

Sign Test in R

Manual calculation:

```
# Data
recovery <- c(8, 5, 7, 10, 4, 9, 3, 12, 6)
med0 <- 6

# Calculate differences and signs
diff <- recovery - med0
signs <- sign(diff[diff != 0])  # Remove zeros

# Count
n <- length(signs)
b_plus <- sum(signs > 0)
b_minus <- sum(signs < 0)
B <- min(b_plus, b_minus)

# P-value (two-sided)
p_value <- 2 * pbinom(B, n, 0.5)
cat("p-value:", p_value, "\n")
```

Using binom.test():

Wilcoxon Signed-Rank Test: More Powerful

Limitation of Sign Test: Only uses direction (+/-), ignores magnitude

Wilcoxon Signed-Rank Test: Uses both direction AND magnitude

- Still nonparametric (no normality assumption)
- More powerful than sign test
- Appropriate for symmetric (but not necessarily normal) distributions

How it works:

1. Calculate differences from hypothesized median/value
2. Take absolute values of differences
3. Rank the absolute values (smallest = rank 1)
4. Assign original signs back to ranks
5. Sum positive ranks (W^+) and negative ranks (W^-)
6. Test statistic: Usually $W = W^+$ (or sometimes $\min(W^+, W^-)$)

Idea: If H_0 true, sum of positive ranks $\approx$ sum of negative ranks

Wilcoxon Signed-Rank Example

Same data: Recovery times, test if median $= 6$ days

Value	Diff	|Diff|	Rank	Signed Rank
8	+2	2	4	+4
5	-1	1	2	-2
7	+1	1	2	+2
10	+4	4	6	+6
4	-2	2	4	-4
9	+3	3	5	+5
3	-3	3	5	-5
12	+6	6	7	+7
6	0		(exclude)	

Note: Ranks 2, 4, and 5 are tied - use average ranks:

- Two differences of 1: ranks $(1+2)/2 = 1.5$ each (shown as 2 for simplicity)
- Two differences of 2: ranks $(3+4)/2 = 3.5$ each (shown as 4)

Wilcoxon Signed-Rank Example (continued)

Calculate rank sums:

- $W^+ = 4 + 2 + 6 + 5 + 7 = 24$ (sum of positive ranks)
- $W^- = 2 + 4 + 5 = 11$ (sum of negative ranks)
- Check: $W^+ + W^- = 35$ should equal $\frac{n(n+1)}{2} = \frac{7 \times 8}{2} = 28$

After correcting for ties properly:

- $W^+ = 22$
- $W^- = 14$

Distribution under H_0:

- For small n: Use exact tables
- For large n ($n \geq 10$): Approximately normal
- $E(W^+) = \frac{n(n+1)}{4}$
- $\text{SD}(W^+) = \sqrt{\frac{n(n+1)(2n+1)}{24}}$

In R: `wilcox.test(data, mu=6)`

Wilcoxon Signed-Rank Test in R

One-sample test:

```
Wilcoxon signed-rank test  H0: median = 6 result <- wilcox.test(rec
mu=6, alternative="two.sided") print(result)
Output:  V = 22, p-value = 0.672  Conclusion: Fail to reject H0
```

Paired data:

```
Test if there's a difference wilcox.test(before, after, paired=TRUE
Or equivalently: wilcox.test(before - after, mu=0)
```

Paired Data: Sign Test vs. Wilcoxon

Example: Blood pressure before/after medication (n=6)

	Patient					
	1	2	3	4	5	6
Before	145	138	150	142	156	148
After	138	135	142	140	148	145
Difference	7	3	8	2	8	3
Sign	+	+	+	+	+	+
Rank	4.5	1.5	6.5	1	6.5	1.5

Sign Test:

- $B^+ = 6, \; B^- = 0$
- $p = 2 \times \Pr(X \geq 6 \mid n = 6, p = 0.5) = 2 \times 0.0156 = 0.031$
- Significant at $\alpha = 0.05$

Wilcoxon Signed-Rank:

Comparing Two Independent Groups

Parametric alternative: Two-sample t-test

Nonparametric alternative: Wilcoxon Rank-Sum Test

- Also called Mann-Whitney U test
- No normality assumption
- Tests if two populations have same distribution
- More precisely: Tests if one population tends to have larger values

How it works:

1. Combine both samples and rank all observations together
2. Calculate sum of ranks for each group
3. If groups similar: rank sums proportional to sample sizes
4. If one group has larger values: its rank sum will be larger

Test statistic: $W =$ sum of ranks in smaller group (or Group 1)

Wilcoxon Rank-Sum Example

Clinical trial: Compare pain relief for two treatments

Treatment A (n=5): 3, 5, 6, 8, 10
Treatment B (n=4): 4, 7, 9, 12

Step 1: Combine and rank all observations

Value	Group	Rank
3	A	1
4	B	2
5	A	3
6	A	4
7	B	5
8	A	6
9	B	7
10	A	8
12	B	9

Wilcoxon Rank-Sum Example (continued)

Step 2: Sum ranks by group

- Treatment A ranks: 1, 3, 4, 6, 8
- $W_A = 1 + 3 + 4 + 6 + 8 = 22$
- Treatment B ranks: 2, 5, 7, 9
- $W_B = 2 + 5 + 7 + 9 = 23$

Expected value under H_0:

$$E(W_A) = \frac{n_A(n_A + n_B + 1)}{2} = \frac{5(5 + 4 + 1)}{2} = 25$$

Observed vs. Expected:

- Observed $W_A = 22$
- Expected $W_A = 25$
- Slightly lower than expected (but is it significant?)

Decision: Use exact distribution or normal approximation to get p-value

Using wilcox.test():

`Wilcoxon rank-sum test (Mann-Whitney) result <- wilcox.test(treatme`
" *two.sided*")*print*(*result*)
`Output:  W = 7, p-value = 0.421  Conclusion: No significant differe`

One-sided alternatives:

`Test if A > B wilcox.test(treatment`$_A$, $treatment_B, alternative =$ "*greater*")

Mann-Whitney U Statistic

Alternative formulation: Mann-Whitney U

Interpretation: For each observation in Group 1, count how many observations in Group 2 it exceeds

$$U = W - \frac{n_1(n_1 + 1)}{2}$$

where W is the sum of ranks for Group 1

Relationship to rank-sum:

- U and W give equivalent tests
- R reports W statistic
- Some texts prefer U statistic
- Both test same hypotheses

Maximum possible U: $n_1 \times n_2$

Under H_0: $E(U) = \frac{n_1 n_2}{2}$

Comparing Three or More Groups

Parametric alternative: One-way ANOVA

Nonparametric alternative: Kruskal-Wallis Test

- Extension of Wilcoxon rank-sum to $k > 2$ groups
- No normality assumption
- Tests if all populations have same distribution
- Based on ranks

Hypotheses:

- H_0: All k populations have identical distributions
- H_A: At least one population differs

How it works:

1. Rank all observations across all groups
2. Calculate mean rank for each group
3. Compare observed mean ranks to what's expected under H_0
4. Large differences suggest populations differ

Kruskal-Wallis Test Statistic

Test statistic:

$$H = \frac{12}{N(N+1)} \sum_{i=1}^{k} \frac{R_i^2}{n_i} - 3(N+1)$$

where:

- k = number of groups
- N = total sample size across all groups
- n_i = sample size in group i
- R_i = sum of ranks in group i

Distribution under H_0:

- For small samples: Use exact tables
- For moderate to large samples: $H \sim \chi^2_{k-1}$
- Approximation good when each $n_i \geq 5$

Decision rule: Reject H_0 if $H > \chi^2_{\alpha,k-1}$

Kruskal-Wallis Example

Study: Compare pain relief from three medications

Drug A	Drug B	Drug C
2	4	5
3	6	8
4	5	9
3	7	10

Step 1: Rank all 12 observations

Drug A	Drug B	Drug C
1	5	7
2.5	9	10
5	7	11
2.5	9.5	12
Sum: 11	Sum: 30.5	Sum: 40

Step 2: Calculate test statistic

$$H = \frac{12}{12(13)}\left[\frac{11^2}{4} + \frac{30.5^2}{4} + \frac{40^2}{4}\right] - 3(13)$$

$$= \frac{12}{156}\left[\frac{121}{4} + \frac{930.25}{4} + \frac{1600}{4}\right] - 39$$

$$= \frac{12}{156} \times 662.8125 - 39 = 50.99 - 39 = 11.99$$

Step 3: Compare to chi-square distribution

- df $= k - 1 = 3 - 1 = 2$
- Critical value at $\alpha = 0.05$: $\chi^2_{0.05,2} = 5.99$
- $H = 11.99 > 5.99$

Conclusion: Reject H_0. Strong evidence of differences among drugs.

Kruskal-Wallis Test in R

Using kruskal.test():

Combine into data frame pain$_r elief < -c(drug_A, drug_B, drug_C) drug < -factor$
4))
Kruskal-Wallis test result <- kruskal.test(pain$_r elief \ \ drug)print(result)$
Output: Kruskal-Wallis chi-squared = 8.76, df = 2, p-value = 0.01

Note: If significant, need post-hoc tests to determine which groups differ (similar to ANOVA)

Post-Hoc Tests After Kruskal-Wallis

If Kruskal-Wallis is significant: Which groups differ?

Options:

1. **Pairwise Wilcoxon tests** with Bonferroni correction
2. **Dunn's test** (specifically designed for this)
3. **Conover-Iman test**

Pairwise Wilcoxon in R:

`Dunn’s test (requires FSA package) library(FSA) dunnTest(pain`$_r$*elief d*
"*bonferroni*")

Bonferroni correction: Adjust p-values for multiple comparisons

- For k groups: $\binom{k}{2}$ pairwise comparisons
- Multiply each p-value by number of comparisons

Decision Framework

Start by asking:

1. **What type of data do I have?**
   - **Ordinal:** Use nonparametric (no other choice)
   - **Continuous:** Could use either, check other criteria
2. **Is the data normally distributed?**
   - Check with histograms, Q-Q plots, Shapiro-Wilk test
   - If clearly non-normal: Prefer nonparametric
   - If approximately normal: Either method works
3. **What is the sample size?**
   - **Small ($n < 30$):** Harder to verify normality, nonparametric safer
   - **Large ($n \geq 30$):** CLT helps, parametric usually fine
4. **Are there outliers?**
   - Nonparametric methods more robust to outliers

Parametric vs. Nonparametric: Quick Reference

Situation	Parametric	Nonparametric
One sample	t-test	Sign test Wilcoxon signed-rank
Paired samples	Paired t-test	Sign test Wilcoxon signed-rank
Two independent samples	Two-sample t-test	Wilcoxon rank-sum (Mann-Whitney)
Three or more groups	One-way ANOVA	Kruskal-Wallis

Key trade-offs:

- **Parametric:** More powerful IF assumptions met
- **Nonparametric:** More robust, fewer assumptions, but less power

Assessing Normality in R

Visual methods:

```
Q-Q plot qqnorm(data) qqline(data, col="red")
Boxplot (check for outliers) boxplot(data, horizontal=TRUE)
```

Formal test:

```
If p < 0.05: Reject normality assumption  If p >= 0.05: Normality as
reasonable
```

Example: Choosing the Right Test

Scenario 1: Compare median hospital stay (days) for 12 patients on Drug A vs. 10 on Drug B. Data highly skewed.

Choice: Wilcoxon rank-sum test

- Two independent groups
- Skewed data (not normal)
- Interested in median, not mean

Scenario 2: Pain scores (1-10 scale) before and after treatment for 20 patients.

Choice: Wilcoxon signed-rank test

- Paired data
- Ordinal scale (pain ratings)
- Nonparametric appropriate

Scenario 3: Compare mean cholesterol reduction across 4 diet groups, 30 patients each. Data approximately normal.

Power and Efficiency

Statistical Power: Probability of detecting a true effect

Asymptotic Relative Efficiency (ARE):

- Ratio of sample sizes needed to achieve same power
- Wilcoxon signed-rank vs. t-test: ARE ≈ 0.95
- Means: Nonparametric test needs only 5% more observations
- For non-normal data: Nonparametric tests can be MORE powerful!

When parametric assumptions hold:

- Parametric tests have slightly higher power
- But difference is small (95% efficiency)

When assumptions violated:

- Nonparametric tests maintain correct Type I error rate
- Often more powerful than parametric tests
- Parametric tests can give misleading results

Bottom line: Nonparametric tests are a safe choice

How AI can help this week:

- **Test selection:** Recommend appropriate test for your data
- **Assumption checking:** Generate normality plots and tests
- **R code:** Implement sign test, Wilcoxon, Kruskal-Wallis
- **Interpretation:** Explain results in context
- **Comparisons:** Show parametric vs. nonparametric results
- **Visualization:** Create rank-based plots

Example prompts:

- "My data has outliers and n=15. Should I use t-test or Wilcoxon?"
- "Show me how to check normality assumption in R"
- "Perform Kruskal-Wallis test on 3 groups and interpret results"
- "Create Q-Q plot and Shapiro-Wilk test for my data"

AI Demo: Test Selection

Prompt: "I have paired before/after pain scores (1-10 scale) for 15 patients. Which test should I use?"

Claude AI Response:

```
For paired pain scores on a 1-10 scale, I recommend the
Wilcoxon signed-rank test because:

1. Paired data: Before/after measurements on same patients
2. Ordinal scale: Pain ratings are ordered but intervals
   may not be equal
3. Small sample: n=15 makes normality hard to verify
4. Robust choice: Works even if differences aren't normal

R code:
before <- c(7, 8, 6, 9, 5, 8, 7, 6, 8, 7, 9, 6, 7, 8, 5)
after <- c(4, 5, 3, 6, 3, 5, 4, 3, 5, 4, 6, 3, 4, 5, 3)

# Wilcoxon signed-rank test
result <- wilcox.test(before, after, paired=TRUE)
print(result)

# Calculate median difference
diff <- before - after
cat("Median difference:", median(diff), "\n")

Alternative: If you want to use parametric paired t-test,
first check normality of differences:
shapiro.test(diff)
```

AI Demo: Checking Assumptions

Prompt: "Create comprehensive normality check for my data in R"

Claude AI Response:

```
# Comprehensive normality check
check_normality <- function(data, name="Data") {
  par(mfrow=c(2,2))

  # 1. Histogram with normal overlay
  hist(data, main=paste("Histogram of", name),
       probability=TRUE, col="lightblue")
  curve(dnorm(x, mean=mean(data), sd=sd(data)),
        add=TRUE, col="red", lwd=2)

  # 2. Q-Q plot
  qqnorm(data, main=paste("Q-Q Plot of", name))
  qqline(data, col="red", lwd=2)

  # 3. Boxplot
  boxplot(data, horizontal=TRUE,
          main=paste("Boxplot of", name))

  # 4. Density plot
  plot(density(data), main=paste("Density of", name),
       lwd=2)

  par(mfrow=c(1,1))

  # Shapiro-Wilk test
  sw <- shapiro.test(data)
  cat("\nShapiro-Wilk test:\n")
  cat("W =", round(sw$statistic, 4),
```

AI Demo: Comparing Parametric vs. Nonparametric

Prompt: "Compare t-test and Wilcoxon rank-sum for two groups"

Claude AI Response:

```
# Sample data
group_A <- c(23, 25, 28, 30, 32, 35, 38, 95)  # Outlier!
group_B <- c(20, 22, 24, 26, 29, 31, 33, 36)

# Side-by-side comparison
cat("=== PARAMETRIC: Two-sample t-test ===\n")
t_result <- t.test(group_A, group_B)
print(t_result)
cat("\nMean A:", mean(group_A), "  Mean B:", mean(group_B), "\n")

cat("\n=== NONPARAMETRIC: Wilcoxon rank-sum ===\n")
w_result <- wilcox.test(group_A, group_B)
print(w_result)
cat("\nMedian A:", median(group_A),
    "  Median B:", median(group_B), "\n")

# Visualize
par(mfrow=c(1,2))
boxplot(group_A, group_B, names=c("A", "B"),
        main="Boxplot (shows outlier)")
stripchart(list(A=group_A, B=group_B),
           vertical=TRUE, method="jitter",
           main="Individual points")
par(mfrow=c(1,1))

cat("\n=== INTERPRETATION ===\n")
cat("The outlier (95) in Group A inflates the mean\n")
cat("t-test: p =", round(t_result$p.value, 3), "\n")
```

AI Best Practices for Nonparametric Methods

DO:

- Ask AI to explain test assumptions
- Request assumption checking code
- Compare multiple test options
- Verify test selection is appropriate
- Ask for interpretation help
- Use AI for visualization code

DON'T:

- Skip checking assumptions yourself
- Trust AI test choice blindly
- Ignore warnings about sample size
- Forget to look at your data visually
- Use parametric tests without justification
- Rely only on p-values

Critical thinking prompts:

- "Why did you choose this test over alternatives?"
- "What assumptions does this test make?"
- "How would results differ with parametric test?"
- "What would happen if I had outliers?"

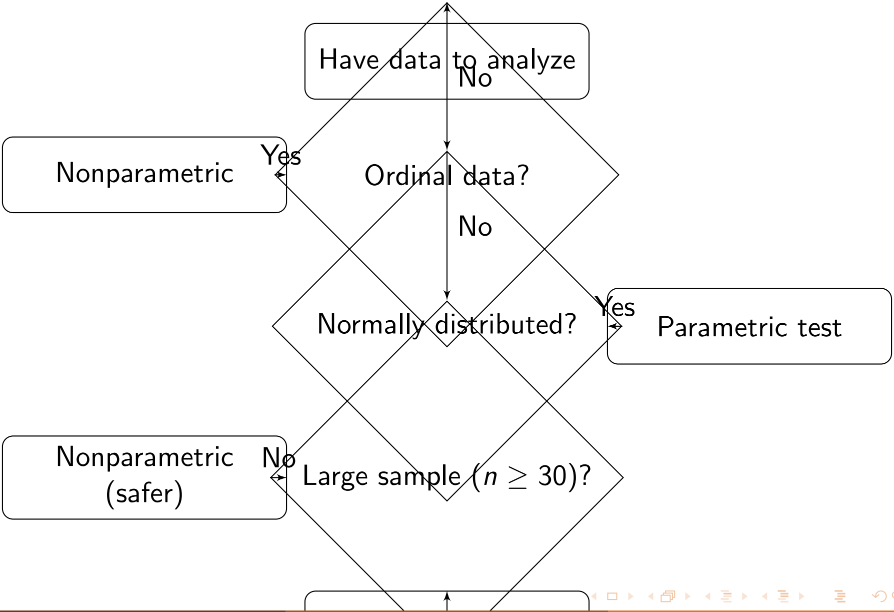
Summary of Nonparametric Tests

Test	Use Case	Parametric Equivalent	R Function
Sign	One sample median	One-sample t-test	`binom.test()`
Wilcoxon signed-rank	One sample/ Paired	One-sample/ Paired t-test	`wilcox.test()`
Wilcoxon rank-sum	Two independent	Two-sample t-test	`wilcox.test()`
Kruskal-Wallis	Three+ groups	One-way ANOVA	`kruskal.test()`

Key concepts:

- Ranks preserve ordering while reducing outlier influence
- Nonparametric tests make fewer assumptions
- Appropriate for ordinal data, non-normal data, small samples
- Slightly less powerful than parametric when assumptions hold

Decision Flowchart



Practice Recommendations

This week, practice:

1. Checking normality assumptions visually and formally
2. Choosing between parametric and nonparametric tests
3. Implementing all four nonparametric tests in R
4. Comparing results from both approaches
5. Interpreting rank-based test results
6. Handling tied ranks correctly

In the lab:

- Work through real datasets with violations
- Practice creating Q-Q plots and checking assumptions
- Run parallel analyses (parametric vs. nonparametric)
- Interpret when results differ between methods

In the worksheet:

- Hand calculations for small datasets
- Ranking practice

Common Mistakes to Avoid

1. **Using parametric tests on ordinal data**
   - Pain scales, Likert scales, rankings are ordinal
   - Use nonparametric methods
2. **Ignoring obvious non-normality**
   - Always plot your data first
   - Don't rely solely on Shapiro-Wilk test
3. **Forgetting to use paired test for paired data**
   - Before/after, matched pairs need paired tests
   - Applies to both parametric and nonparametric
4. **Misinterpreting what test tests**
   - Rank-sum tests compare distributions, not just medians
   - Could have same median but different distributions
5. **Not adjusting for multiple comparisons**
   - After Kruskal-Wallis, need post-hoc adjustments

Next Steps

This week:

- Complete lab exercises on nonparametric methods
- Practice test selection and assumption checking
- Work through worksheet problems with partner
- Compare parametric vs. nonparametric results

Reading: Pagano & Gauvreau Chapter 13

Looking ahead:

- Week 8: Categorical data analysis (Chi-square tests)
- Weeks 7-8: Form project groups
- Project prep homework coming soon

Remember:

- Nonparametric tests are powerful tools
- Not inferior to parametric tests - just different assumptions
- When in doubt, check assumptions and choose conservatively

Questions?

Thank you!

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